CUSTOMER CHURN BUSINESS ANALYSIS

Technical Report | The Developers Arena — Week 8 Capstone

1. Executive Overview

This capstone analyzes 500 customer records to understand churn patterns and convert the findings into business actions. The project follows an end-to-end workflow from data validation through exploratory analysis, statistical testing, customer segmentation, scenario analysis, and implementation planning.

Headline result: 53 of 500 customers churned, producing an overall observed churn rate of 10.60%. The strongest business signals are early tenure and contract type, while monthly charges provide an additional statistically significant differentiator.

2. Business Problem and Objectives

Customer churn can reduce recurring revenue and increase the need to acquire replacement customers. The business objective is to identify customer groups with elevated churn and prioritize retention resources.

Objectives

- Measure the overall churn rate.
- Identify churn differences across contract, tenure, payment, billing, and customer groups.
- Statistically validate important differences and associations.
- Build a simple risk-based customer segmentation.
- Estimate retention scenarios and monthly charge exposure.
- Recommend practical actions and KPIs.

3. Dataset Description

Attribute	Value
Records	500 customers
Columns	9
Target	Churn
Numerical fields	Tenure, MonthlyCharges, TotalCharges, SeniorCitizen, Churn
Categorical fields	Contract, PaymentMethod, PaperlessBilling
Customer identifier	CustomerID

The dataset was validated for missing values, duplicate rows, duplicate customer IDs, categorical values, numeric ranges, binary fields, and customer-ID format. No missing values or duplicate rows were found in the provided dataset, and the cleaned dataset was saved for downstream analysis.

4. Data Preparation and Quality Checks

A dedicated data-cleaning notebook was used to separate preparation from analysis. The raw dataset was loaded, inspected, validated, and saved as cleaned_data.csv. This modular approach improves reproducibility and makes the analytical workflow easier to audit.

Validation checks included:

- Dataset dimensions and data types
- Missing-value counts
- Duplicate-row and duplicate-ID checks
- Categorical-value validation
- Numeric range validation
- Binary-value validation for SeniorCitizen and Churn
- CustomerID pattern validation

5. Exploratory Data Analysis

5.1 Overall Churn

The dataset contains 447 retained customers and 53 churned customers. The observed churn rate is 10.60%, which serves as the baseline for comparing customer segments.

Chart: Figure 1. Overall retained versus churned customers.

5.2 Contract Type

Month-to-month customers have the highest observed churn rate at 20.59%, compared with 4.30% for one-year contracts and 6.94% for two-year contracts. Contract type therefore represents a major retention priority.

Chart: Figure 2. Observed churn rate by contract type.

5.3 Tenure

Customers with 0–12 months of tenure have an observed churn rate of 63.10%. All 53 observed churn cases occur within this group, while the other tenure groups show zero observed churn in this dataset.

Chart: Figure 3. Observed churn rate by tenure group.

5.4 Monthly Charges

Average monthly charges are 111.72 for retained customers and 129.77 for churned customers, a difference of 18.05. The distribution is summarized using a box plot.

Chart: Figure 4. Monthly charges by customer status.

6. Advanced Statistical Analysis

Hypothesis testing was conducted at $\alpha = 0.05$. The tests were selected to validate important differences and associations identified during EDA.

Analysis	Test	Statistic	P-value	Conclusion
Monthly Charges vs Churn	Independent t-test	-2.6464	0.010079	Significant
Tenure vs Churn	Independent t-test	34.5349	<0.001	Significant
Contract Type vs Churn	Chi-square	27.7153	<0.001	Significant
Contract effect size	Cramér's V	0.2354	—	Moderate association

The monthly-charge t-test indicates a statistically significant difference between retained and churned groups ($p = 0.0101$). The tenure t-test shows a very strong difference ($p < 0.001$). The chi-square test indicates a statistically significant association between contract type and churn ($p < 0.001$). Cramér's V of 0.2354 indicates a moderate association. These results establish statistical evidence of differences or association, not causation.

7. Combined Segment Analysis

Combining tenure and contract type provides a more targeted view of risk. The month-to-month + 0–12 month segment contains 35 customers and all 35 are observed as churned, producing a 100% observed churn rate. The one-year + 0–12 month segment has 30.77% churn, while the two-year + 0–12 month segment has 43.48% churn.

Chart: Figure 5. Churn rate by tenure group and contract type.

8. Customer Risk Segmentation

A simple rule-based risk score combines three characteristics: early tenure (≤12 months), high monthly charges (at or above the dataset median), and month-to-month contract. Customers with scores of 3, 2, and 0–1 were classified as high, medium, and low risk respectively.

Risk Level	Customers	Churned	Observed Churn Rate
High Risk	21	21	100.00%
Medium Risk	103	30	29.13%
Low Risk	376	2	0.53%

Chart: Figure 6. Observed churn rate by rule-based customer risk level.

The high-risk segment should be treated as a prioritization framework rather than a production predictive model. Its 100% observed churn rate is a property of this dataset and requires validation on broader real-world data.

9. Retention Scenario Analysis

For the 35 observed churned customers in the highest-priority segment, hypothetical retention scenarios were calculated. A 20% retention rate corresponds to approximately 7 customers retained; 30% corresponds to 10; and 40% corresponds to 14. These are illustrative scenarios, not forecasts.

10. Revenue Exposure Analysis

The 35 customers in the highest-priority month-to-month + 0–12 month segment have an average monthly charge of 120.94 and a combined monthly charge exposure of 4,233.00. The dataset does not specify currency or profitability, so this should be interpreted as charge exposure rather than confirmed company revenue.

11. Business Insights

- Early tenure is the strongest practical retention priority in this dataset.
- Month-to-month customers have substantially higher observed churn than longer-contract customers.
- Higher monthly charges are statistically different between churned and retained customers.
- Payment method, senior-citizen status, and paperless billing show comparatively small churn-rate differences.
- Combining multiple risk indicators enables targeted allocation of retention resources.

12. Recommendations

First-year retention: Implement structured onboarding and 30/60/90-day engagement checkpoints.

Contract conversion: Target suitable month-to-month customers with appropriate loyalty benefits or upgrade incentives.

High-risk outreach: Create a priority queue for customers with multiple risk indicators.

Value review: Review pricing and service-value perception among higher-charge customers.

Measurement: Track churn, retention, contract conversion, campaign response, and customers retained.

13. Implementation Plan

Phase	Action	Primary KPI
1	Identify customer risk segments	Risk coverage
2	Launch early-tenure onboarding	Early churn rate
3	Target month-to-month customers	Contract conversion
4	Proactive high-risk outreach	Retention rate
5	Test personalized offers	Campaign response
6	Review and scale	Monthly churn rate

14. Limitations and Reproducibility

- The analysis uses 500 customer records, so larger samples are desirable.
- All observed churn occurs within 0–12 months, an unusually concentrated pattern that may reflect synthetic/training-data construction.
- Important real-world variables such as satisfaction, support interactions, complaints, and service usage are not available.
- Statistical association does not prove causation.
- The rule-based risk score is not a production machine-learning model.

The project is organized into separate notebooks for cleaning, EDA, and advanced analysis, plus a consolidated capstone notebook. Raw and cleaned data are separated, supporting reproducibility and clearer review.

15. Final Conclusion

The analysis identifies early tenure and contract type as the central churn-related business priorities. The strongest observed risk occurs among month-to-month customers within their first 12 months. A targeted retention program should therefore focus on early engagement, contract conversion, and risk-based outreach, while continuously measuring outcomes. The findings should be validated against larger, representative business data before production deployment.